

Fundamentals of Robot Control — Syllabus

Subject area: Robotics

Format: 7 weeks × (2 lectures + 1 practice day) = **14 lectures**

Style: Breadth over depth — core ideas, simulation, and curated references / GitHub code per topic

Prior course: Full kinematics and dynamics are covered previously; here they are only briefly recalled.

1. Short description

This introductory course surveys robot control from classical methods to optimal and learning-based approaches. Students briefly recall kinematics/dynamics, learn what feedback control is, simulate systems in modern software, design **kinematic (velocity)** and **dynamic (torque)** controllers, and apply optimal control (LQR, trajectory optimization, MPC, MPPI) and learning methods (behavior cloning, DAgger, reinforcement learning).

The final week features **guest lectures from experts in the field** from **Sber Robotics Center** and **Yandex Robotics** (2–3 speakers) on topics **not covered** in the core syllabus.

The course emphasizes intuition, implementation, and method selection over deep proofs. **Advanced methods in both classical and learning-based control are omitted**; Lyapunov theory appears only lightly as energy-based reasoning.

Students also get a **brief, practical introduction** to common robotics software (e.g. MuJoCo, Warp, mjlab, Pinocchio, Newton, Crocoddyl, CasADi, CVXOPT) — enough for labs and take-homes, not a full software course.

2. Intended learning outcomes

Concepts (know / explain)

- What a control system is; feedback vs open-loop; goals of robot control
- Brief recall of kinematics/dynamics as needed for control
- Velocity / Jacobian-based control and linear dynamic control (PD/PID, state feedback)
- Inverse dynamics / gravity compensation; light energy / Lyapunov intuition
- Optimal control: LQR, trajectory optimization, MPC, MPPI
- Behavior cloning; distribution shift; DAgger
- RL: MDP, observations, rewards, policy; locomotion case studies
- When to choose kinematic vs dynamic, classical vs optimal vs learning methods
- Exposure to industrial perspectives on topics beyond the core lectures

Skills (do)

- Simulate robots in MuJoCo and related stacks; wire simple control loops
- Use, at an introductory level, MuJoCo, Warp, mlab, Pinocchio, Newton, Crocoddyl, CasADi, CVXOPT
- Implement velocity- and torque-level controllers on provided plants
- Implement and compare LQR / traj-opt / MPC / MPPI
- Train and evaluate BC, DAgger-style loops, and simple RL setups

Synthesis (apply)

- Select and justify a control stack for a specific robotics problem
- Deliver a course project demonstrating applied understanding

3. Course sections

Section	Weeks	Content	Assessment
I — Intro & classical	1–3	Control intro; brief kin/dyn; simulation; velocity, PD, ID; LQR start	Guided practices (simulate & implement)
II — Optimal control	3–4	LQR, trajectory optimization, MPC, MPPI	Take-home 1 (released start of week 3)
III — Learning	5–6	Week 5: BC + DAgger; week 6: RL + short wrap-up	Take-home 2 (released start of week 5); exam = course project
IV — Industry guests	7	Guest lectures by experts from Sber Robotics Center and Yandex Robotics (topics not covered in class)	Project / Q&A; project presentations as scheduled

4. Lecture schedule (14 lectures)

Week	Theme	Lecture A	Lecture B
1	Intro & simulation	What is control? Feedback, objectives, architecture; brief kinematics & dynamics recap	Simulation: ODEs, MuJoCo / software stack, sensing & actuation
2	Classical control	Velocity / Jacobian-based control (tracking, singularities)	Linear control: PD/PID, full-state feedback
3	Dynamic control & LQR	Inverse dynamics & gravity compensation (+ light Lyapunov / energy)	LQR (+ trajectory-optimization sketch)
4	Receding horizon & sampling	MPC	MPPI
5	Imitation learning	Behavior cloning: setup, data, supervised policies, failure modes	DAgger: algorithm, distribution shift, practice patterns, limits

Week	Theme	Lecture A	Lecture B
6	Reinforcement learning	RL basics: MDP, observations, actions, rewards, policy	Cases: quadruped tracking + rewards; humanoid IL; short course wrap-up
7	Industry guest lectures	Guest talk(s) by experts from Sber Robotics Center / Yandex Robotics	Guest talk(s) continued; Q&A

Trajectory optimization is sketched with LQR (week 3B) and done properly in **Take-home 1** together with **constrained MPC on a quadrotor**. MPC and MPPI each get a full lecture in week 4.

Possible guest lecture topics (2–3 talks by field experts from **Sber Robotics Center** and **Yandex Robotics**; final set depends on speakers):

- Text-to-motion and generative modeling
- VLA, VLM, and foundation models
- State estimation and perception for legged robots
- Joint optimal design and control for complex articulated systems

Each core lecture: idea and key equations → short demo → 2–3 references + 1 GitHub repository.

5. Practice schedule

Practices are held in **MuJoCo** (Python): guided labs, take-home, and the course project all run in the same simulation stack.

Week	Mode	Plant / focus	Short description
1	Guided lab	Simple models → MuJoCo	Write/inspect state-space toys; load a MuJoCo robot; open-loop sim; plot states, actuators, sensors
2	Guided lab	Manipulator	IK + inverse dynamics (one lab): Jacobian / IK tracking; ID / gravity compensation vs PD
3	In-class lab / TH1 start	Cart-pole + quadrotor	LQR in class on a cart-pole and on a quadrotor (hover / tracking); clarify Take-home 1
4	Take-home	Quadrotor	Work on TH1: trajectory optimization + constrained MPC on quadrotor (thrust / rate / box limits); contrast with unconstrained LQR; optional MPPI; checkpoint
5	Take-home	IL task (provided)	Kick off Take-home 2: behavior cloning + DAgger; early progress with TA
6	Take-home / project	RL + project	Reward / RL experiments; course-project progress; TA feedback
7	Project	Course project	Dry-runs / presentations; guest Q&A spillover; final TA feedback

LQR (week 3): in-class on **cart-pole and quadrotor**. **TH1 (weeks 3-4): MPC + traj opt on quadrotor** — same plant as in-class quadrotor LQR so constraints are clearly useful.

Weeks 1–3 are guided / in-class labs. Weeks 4–7 are mainly take-home and project with TA support.

6. Assessment

Take-homes

Released at the **start of each advanced section**. Practice sessions are for questions, TA help, and showing progress.

Take-home	Released	Covers	Intent
TH1 — Optimal control	Start of week 3	Trajectory optimization, constrained MPC (quadrotor); optional MPPI	Plan and control a quadrotor with traj opt + MPC; show why constraints beat unconstrained LQR
TH2 — Learning-based control	Start of week 5	BC, DAgger, RL	Imitation and RL on a control task; analyze failure modes

Suggested due dates: TH1 end of week 4; TH2 end of week 6 or with the project (week 7).

Exam — course project

No classical written exam. Students apply course knowledge to a **specific robotics problem** and present:

1. Problem and model
2. Method choice (kinematic / classical / optimal / learning) with justification
3. Implementation and results
4. Limitations and next steps

The presentation / demo is the exam (typically during or around week 7).

Grade breakdown (draft)

Component	Weight
Practices / participation & progress check-ins	20%
Take-home 1	25%
Take-home 2	25%
Course project (exam)	30%

7. Prerequisites

Required

- Linear algebra (vectors, matrices, eigenvalues)
- Ordinary differential equations / basic dynamical systems
- Python programming (NumPy-level comfort; notebooks and simple debugging)

- Basic Newtonian mechanics (forces, energy, rigid-body intuition)
- **Previous course** covering robot kinematics, dynamics, and modeling / simulation

Helpful (not assumed in depth)

- Basic linear / classical control (PID, feedback intuition)
- Exposure to numerical optimization
- Exposure to supervised machine learning

Kinematics and dynamics are **only briefly reviewed**. Deep prior experience with MPC, RL, or CasADi / MuJoCo is **not** required. This should **not** be a first course in programming or feedback systems.

8. Organization & tools

- Interactive lectures + weekly practice
- Week 7: guest lectures by experts from Sber Robotics Center and Yandex Robotics (2–3 speakers)
- Python; notebooks / local or containerized environments as provided
- Brief hands-on exposure to robotics software, including (as needed by topic):

Dynamics and Simulation / learning envs: MuJoCo, Warp, mjlab, Newton, Pinocchio

Optimal control / traj opt: Crocoddyl, CasADi, CVXOPT

- One running plant across weeks 2–6 where possible so methods are comparable
- Handwritten notes encouraged

9. References (indicative)

Textbooks

- Spong, Hutchinson, Vidyasagar — *Robot Modeling and Control*
- Siciliano, Sciavicco, Villani, Oriolo — *Robotics: Modelling, Planning and Control*
- Lynch & Park — *Modern Robotics: Mechanics, Planning, and Control* ([online](#))
- Murray, Li, Sastry — *A Mathematical Introduction to Robotic Manipulation* ([PDF](#))
- Slotine & Li — *Applied Nonlinear Control*
- Kirk — *Optimal Control Theory: An Introduction*
- Rawlings, Mayne, Diehl — *Model Predictive Control: Theory, Computation, and Design*
- Sutton & Barto — *Reinforcement Learning: An Introduction* ([online](#))
- Brunton & Kutz — *Data-Driven Science and Engineering* (databooku.wisc.edu)

Online courses & lecture series

- [Underactuated Robotics](#) — Russ Tedrake (MIT); notes, videos, code
- [Control Bootcamp](#) — Steve Brunton; linear / nonlinear control
- [Optimal Control and RL CMU 16-745](#) — Zachary Manchester; LQR, traj opt, MPC, RL
- [UC Berkeley CS285 Deep RL](#) — Sergey Levine; imitation learning + RL
- [Stanford CS224R Deep Learnings](#) — RL and learning for robots (optional depth)
- [Slotine Nonlinear Control](#) — nonlinear control lecture series
- [Data-Driven Dynamical Systems & Control](#) — Brunton & Kutz; book + short videos
- [Coursera: Robotics Specialization \(Penn\)](#) — kinematics, dynamics, control refreshers
- [OpenAI Spinning Up in Deep RL](#) — practical RL algorithms overview
- [MuJoCo documentation & tutorials](#) — simulation stack used in labs

Exact paper and GitHub links are provided per lecture.

10. Out of scope (this edition)

Advanced methods in classical and learning-based control are omitted from the taught core (adaptive, robust / H_∞ , sliding mode, full Lyapunov theory, CLF/CBF, advanced RL, foundation / VLA-VLM stacks, generative motion models, etc.). Related themes may appear in **week 7 guest lectures**.

Full kinematics/dynamics derivations belong to the **previous course**. Software coverage is introductory only.